

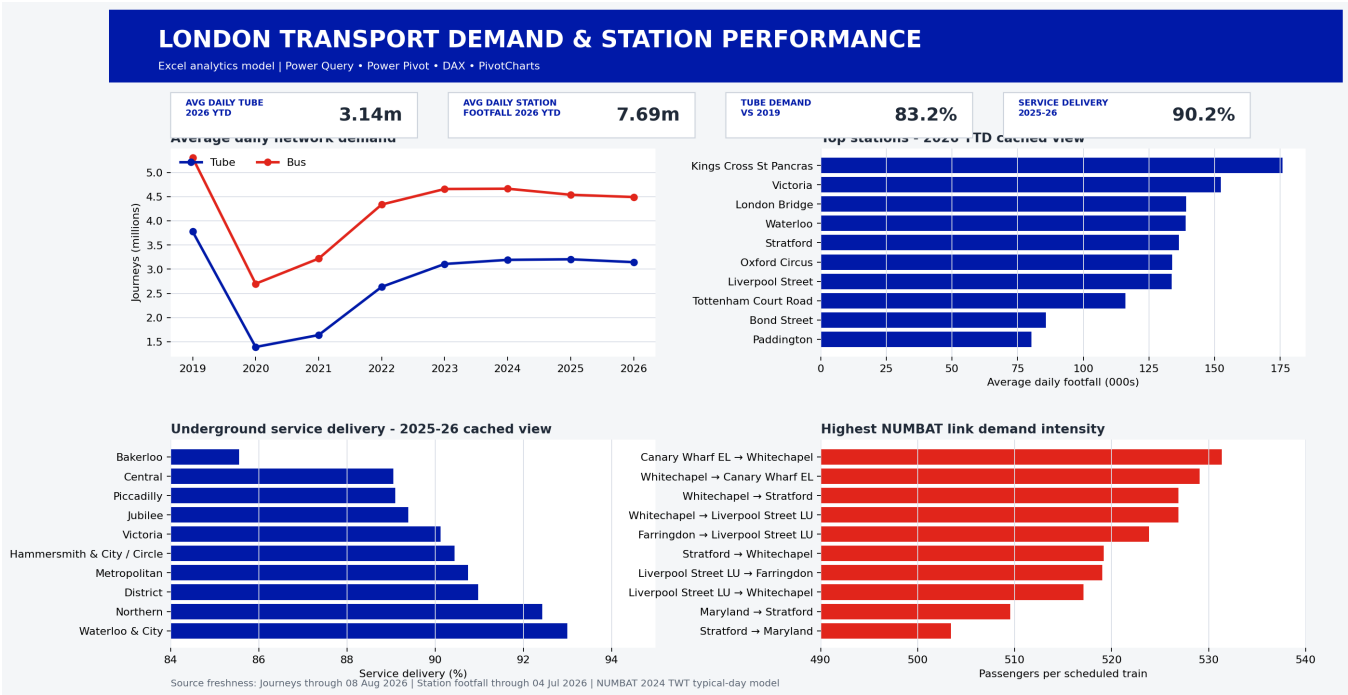
LONDON TRANSPORT DEMAND & STATION PERFORMANCE

Excel analytics model using TfL data | Power Query • Power Pivot • DAX • PivotTables • PivotCharts

Project overview

This workbook analyses London transport demand, station usage, typical 15-minute passenger flows and London Underground service delivery using publicly available Transport for London data. The analytical pipeline is built entirely in Excel: source files are cleaned in Power Query, related in the Power Pivot Data Model, calculated with DAX and surfaced through interactive dashboards and slicers.

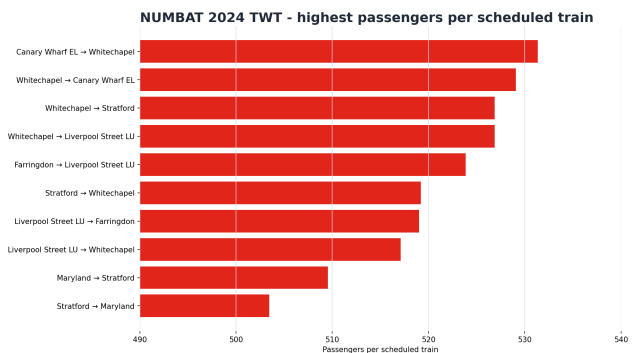
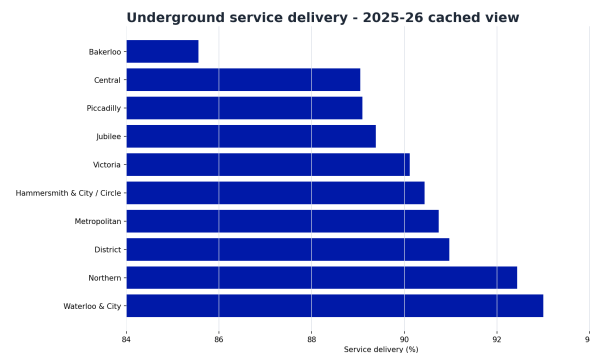
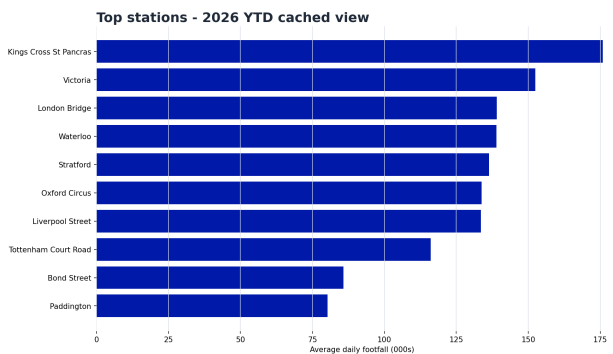
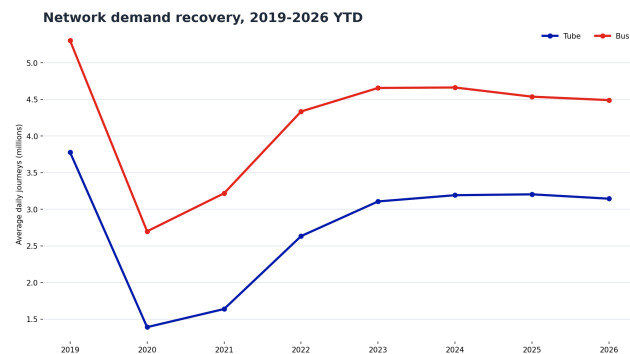
AVG DAILY TUBE 2026 YTD	3.14m	AVG DAILY STATION FOOTFALL 2026 YTD	7.69m
TUBE DEMAND VS 2019	83.2%	SERVICE DELIVERY 2025-26 CACHED VIEW	90.2%



Source freshness: Journey data through 08 Aug 2026; Station Footfall through 04 Jul 2026; performance through 2026/27 YTD; NUMBAT 2024 TWT. Portfolio build refresh metadata: 17 Aug 2026 15:38.

Key findings

- 1. Demand shock and recovery.** Average daily Tube journeys fell from about 3.78m in 2019 to 1.39m in 2020 (-63.2%). Bus demand fell from about 5.30m to 2.70m (-49.1%).
- 2. Recovery remains incomplete.** In 2025, Tube demand was about 84.8% of the 2019 average and Bus demand about 85.6%. The 2026 YTD cached view is about 83.2% and 84.7% respectively.
- 3. Weekday profile.** Thursday has the highest Tube average, Friday the highest Bus average, and Sunday is the lowest-demand day for both.
- 4. 2026 YTD station concentration.** King's Cross St Pancras leads the cached station ranking at roughly 175.9k average daily footfall, ahead of Victoria and London Bridge.
- 5. Line-level variation.** In the cached 2025-26 view, overall service delivery is about 90.2%; Waterloo & City and Northern rank highest, while Bakerloo ranks lowest.
- 6. NUMBAT link intensity.** The all-day cached ranking is led by Canary Wharf EL -> Whitechapel at about 531 passengers per scheduled train. This is not a capacity-utilisation percentage.



Methodology, architecture & limitations

Analytical architecture

Layer	Implementation
Start Here	README sheet and workbook guide
Raw data	External TfL CSV/XLSX files; never manually edited
Power Query	Folder combination, schema standardisation, aliases, data typing, NUMBAT unpivot
Data Model	Date, station, line, quarter-hour and link dimensions filtering fact tables
Calculations	DAX measures and Pivot Support layer
Output	Executive, Network, Station, Line and Peak Flow dashboards
Audit	Data Quality, Methodology and Source Register

Key analytical definitions

- **Average Daily Footfall** = Total Entries + Exits divided by active reporting days.
- **Service Delivery %** = Actual Kilometres / Scheduled Kilometres.
- **Directionality Index** = $\text{ABS}(\text{Entries} - \text{Exits}) / \text{Total Footfall}$.
- **Passengers per Scheduled Train** = NUMBAT Passenger Load / Scheduled Train Frequency.

Data quality

Check	Cached result
Journey rows	2,777
Footfall rows	1,160,147
Distinct station labels	438
Blank station rows	0
Negative footfall rows	0
NUMBAT gate rows	45,216
NUMBAT platform rows	223,872
Non-standard performance rows	50 (expected warning)

Important limitations

- Station Footfall and NUMBAT station entities are not assumed to be identical where TfL naming/aggregation is ambiguous.
- NUMBAT 2024 TWT is a typical autumn Tuesday/Wednesday/Thursday profile, not an annual daily total.
- Passengers per Scheduled Train is a demand-intensity metric, not occupancy or capacity utilisation.
- 2024/25 Underground performance includes partial P6/P8 results and no P7 due to the TfL cyber incident.
- Current-year source families are partial and use different latest dates; source-specific freshness is shown separately.
- The portfolio workbook contains cached data for review; refresh requires the original local TfL source pack and Power Query paths.

Portfolio note: the cached workbook is intended to open for review without recreating the author's local source-folder structure. Data provided by Transport for London.